Fluorescence Spectroscopy: Protein Unfolding

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Intrinsic fluorescence of BSA and lysozyme was measured at 280 nm excitation in PBS and at pH 1. In acid, the BSA emission intensity fell by (23.6 ± 0.4)% and its maximum moved from 346 to 327 nm, a blue shift of (19 ± 2) nm. Lysozyme lost (28.0 ± 0.4)% of its intensity with no measurable shift, (1 ± 2) nm. The uncertainties contain spectral noise only, since each solution was prepared once. A blue shift rather than a red shift indicates that acid rearranges BSA without exposing its Trp residues to water; the shift may also partly reflect a larger tyrosine share of the emission. Together with circular dichroism of solutions prepared the same way, the data indicate a substantial structural change in BSA and little change in lysozyme.

1. INTRODUCTION

Proteins fold into specific three-dimensional shapes that determine their biological function. The native fold is held together by hydrogen bonds, electrostatic interactions, hydrophobic packing and, in some proteins, covalent disulfide bridges. Lowering the pH disrupts this balance by protonating charged side chains, which can partly or completely unfold the protein and expose residues previously buried in its core [1].

The intrinsic fluorophores tryptophan (Trp) and tyrosine (Tyr) respond to their local environment, so their emission reports on tertiary structure without extrinsic labels [2,3]. Comparing emission spectra before and after a perturbation shows whether these residues changed environment.

This report compares the acid response of two well-characterised proteins. Bovine serum albumin (BSA) is a large ($\approx$ 66 kDa), predominantly α-helical transport protein with two Trp residues and 17 disulfide bonds [4]. Hen egg-white lysozyme is smaller ($\approx$ 14 kDa), with six Trp residues and an α-helical and a β-sheet domain joined by four disulfide bonds [5]. Each was measured in PBS and at pH 1.

2. THEORY

2.1 The Fluorescence Process

Fluorescence arises from a three-stage electronic process [2,3]. First, a photon of energy $h\nu_{\mathrm{ex}}$ is absorbed, promoting the fluorophore from the ground singlet state S_0 to a vibrationally excited level of the first excited singlet state, $S_{1'}$. The molecule then relaxes non-radiatively to the lowest vibrational level of S_1 within picoseconds. Finally, it returns to S_0 by emitting a photon of energy $h\nu_{\mathrm{em}}$. Because energy was lost during relaxation, the emitted photon has a longer wavelength than the excitation photon. This difference, the Stokes shift, allows emission to be detected separately from the excitation light.

Not every excited molecule returns to the ground state by fluorescence. Collisional quenching, intersystem crossing and resonance energy transfer can depopulate S_1 without emission. The fraction of absorbed photons that produce fluorescence is the quantum yield [3],

$$\Phi = \frac{k_r}{k_r + \sum_i k_{nr,i}}, \tag{1}$$

where k_r is the radiative rate constant and $k_{nr,i}$ are the rate constants of the non-radiative channels. Both Φ and the emission wavelength depend on the fluorophore's surroundings, which is what makes fluorescence a structural probe.

2.2 Intrinsic Protein Fluorescence

Of the aromatic amino acids, phenylalanine has too low a quantum yield to be useful [2]. Tyr emits near 303 nm, and its maximum hardly depends on environment. The Trp maximum ranges from about 308 nm in a strongly apolar site to about 350 nm in water [3]. Excitation at 280 nm excites both residues; Trp usually dominates the emission of proteins that contain it, partly because Tyr transfers its excitation energy to nearby Trp. Excitation at 295 nm excites Trp alone [2].

Two observables follow. The emission maximum $\lambda_{\max}$ reflects the polarity around the emitting residues: a red shift indicates increased solvent exposure of Trp, and a blue shift a less polar environment or a larger Tyr share of the emission. The peak intensity reflects the quantum yield, which falls when quenchers such as water, protonated histidines, carboxylic acids or disulfide bonds come close to the fluorophore [3].

When a protein unfolds, buried Trp typically becomes exposed: the emission red-shifts towards 350 nm and usually weakens [2]. Partly folded states, such as molten globules or acid-expanded forms, can give more complex changes.

2.3 Fluorescence Intensity in Dilute Solution

The fluorescence intensity F is proportional to the absorbed light and the quantum yield [2],

$$F = I_0 \Phi \left(1 - 10^{-\varepsilon c l}\right) \approx 2.303\, I_0 \Phi \varepsilon c l, \tag{2}$$

where I_0 is the excitation intensity, ε the extinction coefficient, c the concentration and l the path length. The approximation is good to about 10% for absorbance $\varepsilon c l < 0.1$. Computed from the Trp, Tyr and cystine content [6], the 280 nm absorbance of a 0.1 mg/mL solution in a 1 cm cuvette is about 0.07 for BSA but about 0.27 for lysozyme. Lysozyme is therefore outside the linear regime, and its absolute intensity cannot be compared directly with that of BSA. Comparing the same protein in PBS and in acid is unaffected, provided acid does not change its absorbance.

3. METHODS

3.1 Sample Preparation

Four solutions were prepared in Eppendorf tubes, following the manual [2]. Each BSA solution contained 25 µL of 12 mg/mL BSA stock in 2975 µL of solvent, and each lysozyme solution 100 µL of 3 mg/mL lysozyme stock in 2900 µL. The solvent was 1×PBS for the native samples and 0.1 M HCl/KCl at pH 1 for the acid samples.

Each protein was thus diluted to 0.1 mg/mL. One sample of each solution was prepared and measured once.

3.2 Instrument and Measurement

Spectra were recorded on a Cary Eclipse spectrofluorimeter (Agilent, formerly Varian) with the settings in Table I, which match those specified in the manual [2]. The spectra were recorded without the instrument's emission correction.

TABLE I. Cary Eclipse settings for the emission scans.

Parameter	Value
Excitation wavelength	280 nm
Emission range	300–400 nm
Excitation, emission slits	5 nm, 5 nm
Scan rate	600 nm/min
Data interval	1.0 nm
Averaging time	0.1 s
PMT voltage	Medium
Excitation, emission filters	Auto, open
Corrected spectra	Off

Before each scan the quartz cuvette was rinsed with deionised water, ethanol and deionised water, then dried. The instrument reading was zeroed with 1×PBS in the cuvette; no buffer spectrum was recorded or subtracted. Each sample was then scanned, and the spectrum was exported as CSV.

3.3 Analysis

Each spectrum was smoothed with a seven-point running average, and $\lambda_{\max}$ and the peak intensity were taken at its maximum. The instrument software's own peak readout on the unsmoothed spectra agrees with these positions to within 3 nm. The noise was estimated from the scatter of each spectrum about a local quadratic through five neighbouring points, giving 0.9–1.6 intensity units. Adding noise of this size to the smoothed spectrum 4000 times and repeating the peak search gives the uncertainty of $\lambda_{\max}$, 0.9–1.6 nm, and of the peak intensity, about 0.4%. These uncertainties contain spectral noise only.

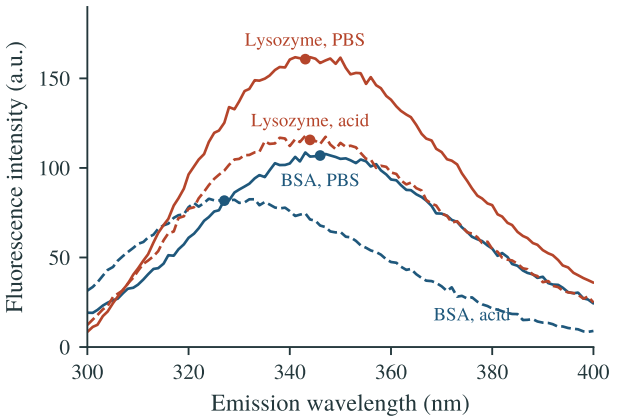


FIG. 1. Fluorescence emission spectra (excitation 280 nm) of BSA and lysozyme in PBS (native) and in 0.1 M HCl/KCl at pH 1 (acid). Colour identifies the protein (BSA blue, lysozyme rust) and line style the solvent (solid native, dashed acid); dots mark the maxima of the smoothed spectra.

4. RESULTS

Fig. 1 shows the four emission spectra, and Table II lists their maxima. Both native spectra peak between 340 and 350 nm. In acid, both proteins emit less. The BSA band moves to shorter wavelength, whereas the lysozyme band stays in place.

TABLE II. Emission maximum $\lambda_{\max}$ (nm) and peak intensity $I_{\max}$ (arbitrary units) of the smoothed spectra in Fig. 1, for BSA and lysozyme (Lys.). $\Delta\lambda$ (nm) and ΔI (%) are the changes relative to the native sample. The uncertainty in the last digit is given in parentheses and contains spectral noise only.

Sample	$\lambda_{\max}$	$\Delta\lambda$	$I_{\max}$	ΔI
BSA native	346(1)	–	106.9(4)	–
BSA + acid	327(1)	−19(2)	81.7(3)	−23.6(4)
Lys. native	343(1)	–	160.7(5)	–
Lys. + acid	344(2)	1(2)	115.7(5)	−28.0(4)

5. ANALYSIS

5.1 BSA: Intensity Decrease with Blue Shift

On acidification the BSA peak intensity falls by (23.6 ± 0.4) %, and $\lambda_{\max}$ moves from 346 to 327 nm, a blue shift of (19 ± 2) nm. The intensity loss is consistent with additional quenching. The blue shift is the opposite of what full unfolding would produce, since Trp moving from the protein interior into water should red-shift towards 350 nm.

BSA therefore does not appear to become a solvent-exposed random coil at pH 1. Between pH 7 and pH 3 BSA converts from the native N form through the F form to the expanded E form, which retains much of its structure because the 17 disulfide bonds hold the backbone together [4]. Two explanations of the blue shift fit the data. Either the two Trp residues sit in less polar surroundings in the acid form, or the Trp emission is quenched more strongly than the Tyr emission, so that Tyr, which emits near 303 nm, makes a larger share of the signal.

The acid spectrum supports a growing Tyr share: at 305 nm it is about 45 units, compared with about 26 for native BSA, although its maximum is lower. The band width is unchanged: the region above 95% of the maximum spans 16 nm in both spectra. These data cannot separate the two explanations.

5.2 Lysozyme: Intensity Decrease without Wavelength Shift

The lysozyme peak intensity falls by $(28.0 \pm 0.4)\%$, while $\lambda_{\max}$ changes by only (1 ± 2) nm, consistent with no shift. The average polarity around the emitting Trp residues therefore appears unchanged. A lower intensity with an unchanged $\lambda_{\max}$ is consistent with a largely native fold in which protonation of groups near the Trp residues makes quenching more effective. The data do not show whether the tertiary structure has loosened.

5.3 Comparison with Circular Dichroism

Solutions prepared the same way were also measured by circular dichroism. At pH 1, the BSA α-helix bands at 210 and 222 nm weakened by 21% and 23%, whereas the lysozyme minimum changed by $(1.0 \pm 0.7)\%$ and its 222 nm band by $(7.3 \pm 0.7)\%$. The two techniques agree: BSA changes substantially in both secondary and Trp environment, while lysozyme changes little in either. Its intensity loss alone does not establish unfolding.

6. DISCUSSION

Both proteins lose a quarter of their emission intensity in acid, so the size of the intensity change does not distinguish them. The wavelength does: BSA shifts by 19 nm, lysozyme not measurably. With the CD result, the data are consistent with acid rearranging BSA, while lysozyme stays close to native. Its four disulfide bonds and compact fold [5] are a plausible reason, but this experiment does not test it, since the two proteins differ in size, sequence and Trp content as well as in disulfide bonding.

The quoted uncertainties are smaller than the true ones, because several contributions could not be assessed. First, each solution was prepared and measured once, so pipetting and concentration differences between paired samples are unknown. They scale ΔI directly but do not move $\lambda_{\max}$. Second, the lysozyme absorbance of about 0.27 attenuates the excitation within the cuvette; this affects comparisons between the proteins, not the acid change of one protein. Third, the spectra were not corrected for the wavelength dependence of the detection system, so absolute $\lambda_{\max}$ values may be offset by a few nanometres; the shifts between spectra are affected much less. Fourth, no buffer spectrum was subtracted, and the acid buffer was never measured on its own. Water Raman scattering near 310 nm adds a small, nearly equal background to all four spectra.

7. CONCLUSION

At 280 nm excitation, acid at pH 1 lowered the emission intensity of BSA by $(23.6 \pm 0.4)\%$ and blue-shifted its maximum by (19 ± 2) nm, while lysozyme lost $(28.0 \pm 0.4)\%$ of its intensity with no measurable shift. Together with the CD measurements, this indicates a substantial structural change in BSA, consistent with its acid-expanded form, and little change in lysozyme. Four improvements would make the results more conclusive. Repeat preparations of each solution would give an uncertainty on ΔI that includes pipetting. Excitation at 295 nm would isolate Trp and test whether the BSA blue shift comes from Trp or from Tyr. Recording the PBS and acid buffers would allow background subtraction, and using corrected spectra would give absolute $\lambda_{\max}$ values. Diluting lysozyme to an absorbance below 0.1 would allow a direct intensity comparison between the two proteins.

REFERENCES

[1] D. Sheehan, *Physical Biochemistry: Principles and Applications* (Wiley, 2000).

[2] PC3267 Biophysics II, *Fluorescence Spectroscopy: A Tool for Protein Folding/unfolding Study* (Laboratory manual, National University of Singapore, n.d.).

[3] J. R. Lakowicz, *Principles of Fluorescence Spectroscopy*, 3rd ed. (Springer, 2006).

[4] T. Peters Jr., *All About Albumin: Biochemistry, Genetics, And Medical Applications* (Academic Press, San Diego, 1996).

[5] C. C. F. Blake, D. F. Koenig, G. A. Mair, A. C. T. North, D. C. Phillips, and V. R. Sarma, Structure of Hen Egg-White Lysozyme: A Three-dimensional Fourier Synthesis at 2 Å Resolution, Nature **206**, 757 (1965).

[6] C. N. Pace, F. Vajdos, L. Fee, G. Grimsley, and T. Gray, How to measure and predict the molar absorption coefficient of a protein, Protein Science **4**, 2411 (1995).